PASCAL'S TRIANGLE MOD 2 AND MOD 3

From binomials to bits and trits

Colour the remainders: the number base becomes visible as geometry.

1 Expand, add, reduce

$$(1+x)^n = \sum_{k=0}^{n} \binom{n}{k} x^k, \qquad (1+x)^4 = 1 + 4x + 6x^2 + 4x^3 + x^4.$$

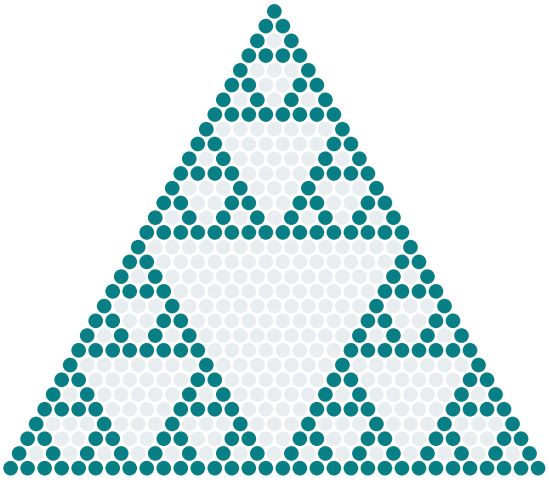
Row n contains the coefficients $\binom{n}{k}$. Replace each coefficient by its remainder $R_p(n,k)$:

$$R_p(n,k) = \binom{n}{k} \bmod p, \qquad R_p(n,k) = \big[R_p(n-1,k-1) + R_p(n-1,k)\big] \bmod p.$$

Start with $R_p(0,0) = 1$ and zeros outside the triangle. Each entry adds its two parents, then keeps only the remainder.

2a Mod 2: Sierpiński's triangle

$2^5 = 32$ rows: $0 \le n < 32$.

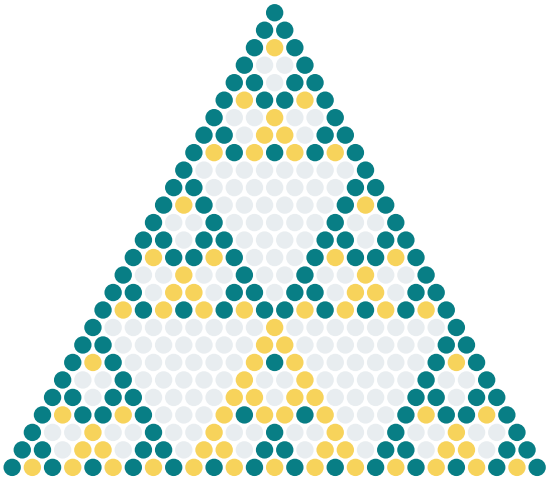


0: even 1: odd

The seed is $1;\ 1,1$. Three copies repeat at scale $1/2$. Parent addition mod 2 is the bit operation XOR.

2b Mod 3: a ternary cousin

$3^3 = 27$ rows: $0 \le n < 27$.



0 1 2

The seed is $1;\ 1,1;\ 1,2,1$. Six copies repeat at scale $1/3$. The central copy multiplies by 2, swapping colours 1 and 2.

3 Lucas's theorem: read the digits

For prime p, write $n = \sum_i n_i p^i$ and $k = \sum_i k_i p^i$, with $0 \le n_i, k_i < p$. Pad with leading zeros.

$$\boxed{\binom{n}{k} \equiv \prod_i \binom{n_i}{k_i} \pmod p}$$

Use $\binom{a}{b} = 0$ when $b > a$. An entry is nonzero precisely when **every digit of k fits beneath the corresponding digit of n.**

4 One coefficient, two readings: $\binom{14}{4} = 1001$

Binary: bits

$$14 = (1110)_2, \qquad 4 = (0100)_2.$$

$$\binom{14}{4} \equiv \binom{1}{0}\binom{1}{1}\binom{1}{0}\binom{0}{0} = 1 \pmod 2.$$

Every 1-bit of k fits: remainder 1.

Ternary: trits

$$14 = (112)_3, \qquad 4 = (011)_3.$$

$$\binom{14}{4} \equiv \binom{1}{0}\binom{1}{1}\binom{2}{1} = 2 \pmod 3.$$

The factor $\binom{2}{1} = 2$ gives remainder 2.

5 Why the base controls the pattern

For prime p, the middle coefficients of $(1+x)^p$ are divisible by p, so $(1+x)^p \equiv 1 + x^p \pmod p$. Consequently,

$$(1+x)^n = \prod_i \big[(1+x)^{p^i}\big]^{n_i} \equiv \prod_i (1+x^{p^i})^{n_i} \pmod p.$$

Choosing $x^{k_i p^i}$ from each factor gives Lucas's product. Each extra digit repeats the seed at a new scale: **the same bits and trits that encode the indices also determine the colours and the gaps.**

Lucas's theorem: R. Meštrović (2014), arXiv:1409.3820. Self-similarity: R. P. Kubelka (2004), *Fibonacci Quarterly* 42(1), 70–75.